SRI VENKATESWARA COLLEGE OF ENGINEERING
QUESTION BANK — DATA STRUCTURES AND ALGORITHMS

Subject Code: CS3301 | Semester: III | Regulation: R2021

UNIT 1 — LINEAR DATA STRUCTURES

Stack, Queue, Arrays, Lists

PART - A (2 Marks)

1.	Define data structure and classify its types.	[2]	K1
2.	State the difference between linear and non-linear data structures.	[2]	K1
3.	What is a stack? Mention its basic operations.	[2]	K1
4.	List the applications of stack in real-time systems.	[2]	K1
5.	Define a queue. Mention the types of queues.	[2]	K1
6.	What is a deque? List its operations.	[2]	K1
7.	Define a linked list. State its advantages over arrays.	[2]	K1
8.	Mention the types of linked lists.	[2]	K1
9.	Write the postfix notation for the expression (A+B)*(C-D).	[2]	K2
10.	Convert the infix expression A+B*C-D to postfix form.	[2]	K2
11.	What is overflow and underflow in stack?	[2]	K1
12.	State the applications of queue in operating systems.	[2]	K1
13.	Define circular queue and mention its advantage.	[2]	K1
14.	What is a doubly linked list? State its operations.	[2]	K1
15.	Mention the difference between array and linked list.	[2]	K1

PART - B (16 Marks)

16.	Explain the array representation of stack and queue with suitable examples. Discuss push, pop, enqueue, and dequeue operations with algorithms.	[16]	K2
17.	Describe the singly linked list. Write algorithms for insertion at beginning, middle, and end. Also write deletion algorithm.	[16]	K2
18.	Explain doubly linked list with its insertion and deletion algorithms. Compare it with singly linked list.	[16]	K2
19.	Explain circular queue. Write algorithms for insertion and deletion operations in circular queue.	[16]	K2
20.	Discuss the applications of stack. Write algorithms for infix to postfix conversion and postfix expression evaluation.	[16]	K2

UNIT 2 — SORTING AND SEARCHING

Sorting Algorithms, Searching Techniques

PART - A (2 Marks)

21.	Define sorting. Classify sorting algorithms.	[2]	K1
22.	What is bubble sort? State its time complexity.	[2]	K1
23.	Define selection sort and mention its advantage.	[2]	K1
24.	What is insertion sort? State its best and worst case complexity.	[2]	K1
25.	Define merge sort. What strategy does it use?	[2]	K1
26.	What is quick sort? What is the role of pivot?	[2]	K2
27.	State the time complexity of binary search.	[2]	K1

28.	What is linear search? When is it preferred over binary search?	[2]	K1
29.	Define stable sorting. Give one example.	[2]	K1
30.	What is the difference between internal and external sorting?	[2]	K2

PART - B (16 Marks)

31.	Analyze the time and space complexity of Bubble Sort, Selection Sort, Insertion Sort, and Merge Sort. Compare their performance.	[16]	K4
32.	Explain merge sort with an example. Trace the execution for: 38, 27, 43, 3, 9, 82, 10.	[16]	K2
33.	Explain quick sort algorithm. Trace the execution for: 10, 80, 30, 90, 40, 50, 70.	[16]	K2
34.	Compare linear search and binary search with algorithms. Analyze time complexity of binary search.	[16]	K4
35.	Discuss the divide and conquer strategy. Apply it to merge sort and analyze its recurrence relation.	[16]	K4

UNIT 3 — TREES

Binary Trees, BST, AVL Trees, Heaps

PART - A (2 Marks)

36.	State the properties of a binary tree.	[2]	K1
37.	Define BST. State the BST property.	[2]	K1
38.	What is a complete binary tree?	[2]	K1
39.	Define AVL tree. What is balance factor?	[2]	K1
40.	What is a heap? Mention its types.	[2]	K1
41.	Define tree height and depth.	[2]	K1
42.	State the difference between full binary tree and complete binary tree.	[2]	K1
43.	What are tree traversal methods?	[2]	K1
44.	Define expression tree.	[2]	K1
45.	What is a min-heap? Give an example.	[2]	K1

PART - B (16 Marks)

46.	Explain binary search tree (BST). Construct a BST for: 50, 30, 70, 20, 40, 60, 80. Perform in-order, pre-order, and post-order traversal.	[16]	K2
47.	Discuss AVL trees and their rotations. Explain LL, RR, LR, RL rotations with diagrams. Construct AVL tree for: 10, 20, 30, 40, 50, 25.	[16]	K2
48.	Explain heap sort. Construct a max-heap for: 4, 10, 3, 5, 1. Trace the sorting process.	[16]	K3
49.	Explain binary tree traversal: in-order, pre-order, post-order. Write recursive algorithms. Apply on a sample tree.	[16]	K2
50.	Discuss the construction of an expression tree. Evaluate the expression tree for: (A+B)*(C-D/E).	[16]	K3

UNIT 4 — GRAPHS

Graph Representation, BFS, DFS, Spanning Trees

PART - A (2 Marks)

51.	Define graph and mention its types.	[2]	K1
52.	What is an adjacency matrix? Give an example.	[2]	K1
53.	Define BFS traversal.	[2]	K1
54.	What is DFS traversal?	[2]	K1
55.	Mention the difference between BFS and DFS traversal.	[2]	K2

56.	Define spanning tree.	[2]	K1
57.	What is a minimum spanning tree (MST)?	[2]	K1
58.	State Kruskal's algorithm for MST.	[2]	K1
59.	Define topological sorting.	[2]	K1
60.	What is a directed acyclic graph (DAG)?	[2]	K1

PART - B (16 Marks)

61.	Compare BFS and DFS graph traversal. Apply both on graph V={A,B,C,D,E}, E={AB,AC,BD,BE,CD,CE,DE}. Draw traversal order.	[16]	K4
62.	Explain Prim's and Kruskal's algorithms for MST. Apply Kruskal's algorithm to find MST of a given weighted graph.	[16]	K3
63.	Explain Dijkstra's shortest path algorithm. Apply it to find shortest paths from source vertex A.	[16]	K3
64.	Discuss graph representations: adjacency matrix and adjacency list. Compare their time and space complexity.	[16]	K4
65.	Explain topological sorting using DFS. Apply it on a given DAG and find the topological order.	[16]	K3

UNIT 5 — HASHING

Hash Functions, Collision Resolution

PART - A (2 Marks)

66.	What is hashing? List any two collision resolution techniques.	[2]	K1
67.	Define hash function. Give an example.	[2]	K1
68.	What is a collision? How is it handled?	[2]	K1
69.	Define open addressing. Mention its types.	[2]	K1
70.	What is chaining? When is it preferred?	[2]	K1
71.	Define load factor in hashing.	[2]	K1
72.	What is linear probing? Mention its disadvantage.	[2]	K1
73.	Define quadratic probing.	[2]	K1
74.	What is double hashing? Give an example.	[2]	K1
75.	State the properties of a good hash function.	[2]	K1

PART - B (16 Marks)

76.	Design a hash table of size 10 using chaining for keys: 25, 42, 96, 101, 102, 162, 197. Analyze performance and compare with open addressing.	[16]	K6
77.	Explain linear probing. Insert keys 10, 22, 31, 4, 15, 28, 17, 88, 59 into a hash table of size 11 using linear probing.	[16]	K3
78.	Compare open addressing techniques: linear probing, quadratic probing, and double hashing with examples.	[16]	K4
79.	Discuss the properties of a good hash function. Explain division method, multiplication method, and folding method.	[16]	K2
80.	Explain extendible hashing. Compare it with static hashing. Discuss advantages and limitations.	[16]	K4

*** END OF QUESTION BANK ***